Acetylsalicylic Acid (ASA)

Objective:

To provide information on the use of acetylsalicylic acid in the prevention of vascular thromboembolic events.

Background:

Acetylsalicylic acid (ASA) is well-established in the management of acute myocardial infarction and in the secondary prevention of cardiovascular disease among both men and women, based on large randomized trials. ASA may also be considered in selected patients for prevention of recurrent venous thromboembolism (VTE) and for prevention of VTE following joint arthroplasty. In addition, in select patients with subclinical vascular disease, ASA may be considered for primary prevention of cardiovascular events. The net benefit of ASA use depends on weighing the anticipated reduction in cardiovascular events against the increased risk of gastrointestinal and intracranial bleeding.

Mechanism of Action:

ASA irreversibly inhibits platelet aggregation by inhibiting thromboxane A_2 (TxA_2) synthesis.

Indications:

(Unless otherwise indicated, ASA refers to enteric-coated ASA)

Antiplatelet therapy with ASA is recommended for all patients with the following, unless contraindicated:

Cardiac: For all patients with symptomatic coronary artery disease (CAD), including:

1. History of acute coronary syndrome (ACS) with and without ST segment elevation
2. Chronic stable angina
3. Following percutaneous coronary intervention (PCI) or coronary artery bypass graft (CABG)
   - In combination with clopidogrel (elective PCI), prasugrel, or ticagrelor (PCI for ACS) while the patient remains at very high risk for subsequent coronary events (usually one year) [see **Clinical Guide: Duration of Dual Antiplatelet Therapy in Patients With Coronary Artery Disease**]
   - In combination with clopidogrel and oral anticoagulation in patients with an indication for anticoagulation, for the shortest possible duration, while the patient is at extremely high risk for subsequent coronary events [see **Clinical Guide: Anticoagulation in Patients Requiring Antiplatelet Therapy**].
4. May be considered in combination with rivaroxaban 2.5 mg twice daily for patients with coronary artery disease or peripheral artery disease for prevention of stroke, myocardial infarction and cardiovascular death, and for the prevention of acute limb ischemia and mortality [see **Clinical Guide: Rivaroxaban (Xarelto®**)
5. Prosthetic heart valves:
   - ASA 81 mg daily is recommended in addition to vitamin K antagonist (VKA) in selected high-risk patients with a mechanical valve prosthesis (see **Clinical Guide: Mechanical and Bioprosthetic Heart Valves: Anticoagulant Therapy**)

- ASA 81 mg daily is reasonable in all patients with a bioprosthetic aortic or mitral val
  an oral anticoagulant (see **Clinical Guide: Mechanical and Bioprosthetic Heart V**
  Anticoagulant Therapy)

Cerebrovascular:

- For patients with history of non-cardioembolic ischemic stroke or transient ischemic attack (TIA) [see **Clinical Guide: Stroke: Secondary Prevention**]
- Following carotid endarterectomy

Peripheral arterial disease (PAD):

- Symptomatic patients with or without a history of vascular intervention [see **Clinical Guide: Peripheral Arterial Disease**]

Primary prevention of cardiovascular morbidity and mortality:

The use of ASA is not routinely recommended for **primary prevention** of a first vascular event.

- This recommendation pertains to individuals with or without vascular risk factors who have **not** had a vascular event.
- The net benefit of ASA in individuals with asymptomatic atherosclerosis (i.e. asymptomatic coronary or carotid atherosclerosis) is uncertain.

Secondary prevention of recurrent VTE:

- In patients at high risk of VTE recurrence following completion of initial anticoagulant therapy for a minimum of 3-6 months, ASA may be considered as an option to decrease recurrent VTE only if extended anticoagulant therapy is not acceptable to the patient. [see **Clinical Guide: Venous Thromboembolism: Duration of Treatment**]

Thromboprophylaxis following joint arthroplasty:

- For patients not at high risk of VTE, consideration can be given to rivaroxaban 10 mg orally per day until post-operative day 5, followed by ASA 81 mg daily for an additional 9 days following total knee arthroplasty or for 30 days after total hip arthroplasty. [see **Clinical Guide: Thromboprophylaxis: Orthopedic Surgery**]

Prevention of Preeclampsia in Pregnancy

- For patients at moderate to high risk of preeclampsia initiated between 12-16 weeks gestation age and discontinued at 36 weeks gestation age.

Primary prevention of cardiovascular morbidity and mortality:

The use of ASA is not routinely recommended for primary prevention of a first vascular event.

- This recommendation pertains to individuals with or without vascular risk factors who have **not** had a vascular event.

- The net benefit of ASA in individuals with asymptomatic atherosclerosis (i.e. asymptomatic atherosclerosis) is uncertain.

Dosing:

- The usual antiplatelet dose of ASA is 81 mg daily. Higher doses are not associated with any evidence of increased efficacy but may be associated with increased bleeding risk.
- A single initial dose of 162 mg chewed or crushed is recommended in patients suffering an ACS event, and then indefinite therapy with 81 mg daily.
- A single initial dose of 325 mg chewed or crushed is recommended in patients prior to angioplasty (if not already receiving chronic daily ASA).
- An initial dose of 81 mg once daily should be utilized in patients suffering a TIA or ischemic stroke of noncardiac origin.

  Note: For patients with a history of, or at high risk of, gastrointestinal bleeding, consider co-administration of a gastroprotective agent (i.e. proton pump inhibitor [PPI], H_2 antagonist) or using an alternate antiplatelet agent such as clopidogrel.

Adverse Effects:

- Bruising and other minor bleeding is very common with use of ASA
- Dyspepsia
- The most common serious adverse effect of ASA is bleeding into the gastrointestinal tract (which is dose-related)
- Allergic reactions, aspirin-induced asthma (may be severe)

Peri-procedural management for patients taking ASA:

Peri-procedural management of ASA is nuanced and requires risk stratification of thrombotic and bleeding risks. See **Clinical Guide: Perioperative Management of Antiplatelet Therapy** for more detailed information, and consultation with a specialist is advised for higher risk patients (e.g. those on dual antiplatelet therapy). Interrupting ASA perioperatively can be considered in those patients who are undergoing a high bleeding risk procedure while also at a low thrombotic risk (ASA should otherwise be continued, especially in patients at high thrombotic risk and/or in those undergoing low bleeding risk procedures.

If interrupting ASA perioperatively, ASA can be stopped 4-7 days prior to the procedure and restarted within 48 hours of surgery (provided there is no significant bleeding).

Special Considerations:

Concomitant use of therapeutic doses of anticoagulants and ASA is discouraged and should only be considered in patients with an indication for anticoagulation who are at relatively low bleeding risk AND have one of the following indications:

- Very recent ACS and/or PCI +/- coronary artery stent (short duration of ASA in combination with clopidogrel and an anticoagulant, after which dual pathway with clopidogrel and an anticoagulant alone is preferred)
- High risk prosthetic heart valve (e.g. older generation valve or mitral valve replacement with atrial fibrillation, left ventricle (LV) dysfunction or previous systemic embolus on therapeutic anticoagulation)
- Proven TIA/ischemic stroke while on therapeutic doses of anticoagulation alone

- High risk thrombophilia with breakthrough thrombosis despite therapeutic anticoagulation (antibody syndrome)

In **most** other patients, ASA should be stopped if anticoagulation is started (See **Clinical Guide: Anticoagulation in Patients Requiring Antiplatelet Therapy**).

Consultation with a thrombosis specialist and/or cardiologist should be considered prior to antiplatelet agent use combined with therapeutic anticoagulation.

Patients taking ASA for vascular protection should avoid the concomitant use of **NSAIDs.** If a patient taking low-dose ASA for vascular protection requires an anti-inflammatory agent, specific cyclooxygenase-2 inhibitors should be chosen over traditional NSAIDs.

ASA should be avoided or used with caution in patients with **asthma or nasal polyps**, in those at **high risk of bleeding** or with recent major bleeding, and in patients with **severe thrombocytopenia** or with familial or **acquired bleeding disorders**.

Pediatrics:

When possible, pediatricians with expertise in thromboembolism should be involved when ASA is being considered for antiplatelet therapy. When this is not possible, a combination of a neonatologist/pediatrician and an adult hematologist, supported by consultation with an experienced pediatric hematologist, is recommended.

Other Relevant

- Mechanical and Bioprosthetic Heart Valves: Anticoagulant Therapy
- Perioperative Management of Antiplatelet Therapy
- Peripheral Arterial Disease
- Stroke: Secondary Prevention
- Thromboprophylaxis: Orthopedic Surgery
- Venous Thromboembolism: Duration of Treatment

Relevant Patient Materials:

ASA Patient information sheet:

References:

Anderson D, et al. Aspirin or rivaroxaban for VTE prophylaxis after hip or knee arthroplasty. N Engl J Med 2018;378:699-707.

Bainey KR, et al. 2023 Canadian Cardiovascular Society/Canadian Association of Interventional Cardiology Focused Update of the guidelines for the use of antiplatelet therapy. Can J Cardiol. 2024;40:160-181.

Bowman W, et al. Effects of Aspirin for primary prevention in persons with diabetes mellitus. N Engl J Med 2018;379:1529-1539.

Duley L, et al. Antiplatelet agents for preventing pre-eclampsia and its complications. Cochrane Database Syst Rev 2019:Oct 30(10).

Douketis, JD. et al. Perioperative management of antithrombotic therapy. CHEST 2022;126(5)

Eikelboom JW, et al. Antiplatelet drugs: antithrombotic therapy and prevention of thrombosis, 9
College of Chest Physicians Evidence-Based Clinical Practice Guidelines. Chest 2012;141(2 9

Eikelboom JW, et al. Rivaroxaban with or without aspirin in stable cardiovascular disease. N Engl J Med 2017;377(14):1319-1330.

Gaziano JM, et al. Use of aspirin to reduce risk of initial vascular events in patients at moderate risk of cardiovascular disease (ARRIVE): a randomised, double-blind, placebo-controlled trial. Lancet. 2018;392(10152):1036-1046.

Lopes RD, et al. Antithrombotic therapy after acute coronary syndrome or PCI in atrial fibrillation. N Engl J Med 2019;380:1509-1524.

Mehta SR, et al. 2018 Canadian Cardiovascular Society/Canadian Association of Interventional Cardiology Focused Update of the guidelines for the use of antiplatelet therapy. Can J Cardiol 2018;34(3):214-233.

Monagle P, et al. Antithrombotic therapy in neonates and children: Antithrombotic therapy and prevention of thrombosis, 9th ed: American College of Chest Physicians Evidence-Based Clinical Practice Guidelines. Chest 2012;141(2 Suppl):e737S-801S.

Nishimura R, et al. 2017 AHA/ACC Focused Update of the 2014 AHA/ACC Guideline for the Management of Patients with Valvular Heart Disease: A report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. Circulation 2017;135(25):e1159-e1195.

Rolnik DL, et al. Aspirin versus placebo in pregnancies at high risk for preterm preeclampsia. N Engl J Med 2017;377(7):613–622.

Wein T, et al, for the Heart and Stroke Foundation of Canada in collaboration with the Canadian Stroke Consortium. Canadian Stroke Best Practice Recommendations, seventh edition: Acetylsalicylic acid for prevention of vascular events. CMAJ 2020 March 23;192:E302-311.

Weitz JI, et al. Rivaroxaban or aspirin for extended treatment of venous thromboembolism. N Engl J Med 2017;376(13):1211-1222.

Yasuda S, et al. Antithrombotic therapy for atrial fibrillation with stable coronary disease. N Engl J Med 2019;381:1103-1113.

Yusuf S, et al. Polypill with or without aspirin in persons without cardiovascular disease. N Engl J Med 2021;384:216-228.

s clinical guides remain current and reliable. Your support helps protect the quality and continuity of these resources for clinicians nationwide.

Date of Version: 21April2025

Please note that the information contained herein is not to be interpreted as an alternative to medical advice from your doctor or other professional healthcare provider. If you have any specific questions ? about any medical matter, you should consult your doctor or other professional healthcare providers, and as such

delay seeking medical advice, disregard medical advice or discontinue medical treatment beca
information contained herein.